



Fundamentals of Deep Learning

Part 3: Convolutional Neural Networks



Agenda

- Part 1: An Introduction to Deep Learning

- Part 2: How a Neural Network Trains

- Part 3: Convolutional Neural Networks
- Part 4: Data Augmentation and Deployment

- Part 5: Pre-Trained Models

- Part 6: Advanced Architectures

Recap of the exercise

Trained a dense neural network model

Training accuracy was high

Validation accuracy was low

Evidence of overfitting

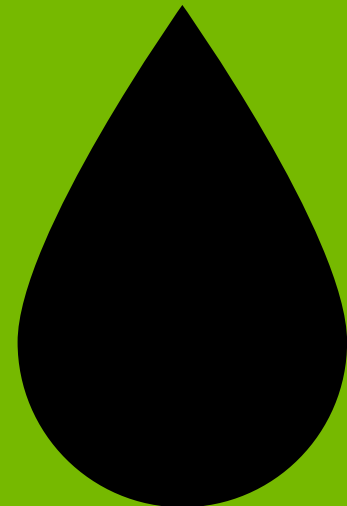


Kernels and Convolution

Kernels and Convolution



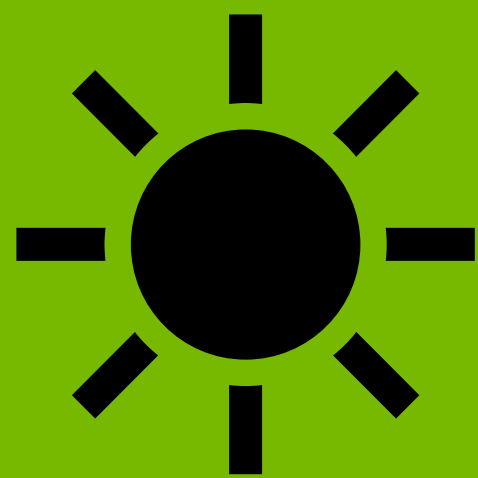
Original Image



Blur



Sharpen



Brighten



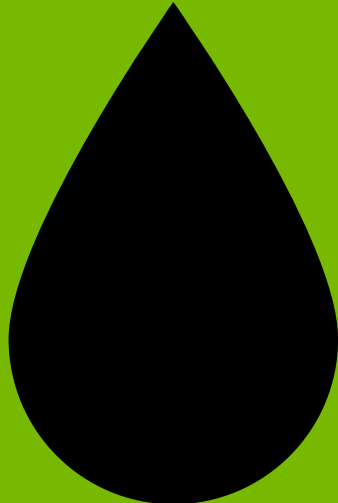
Darken



Kernels and Convolution

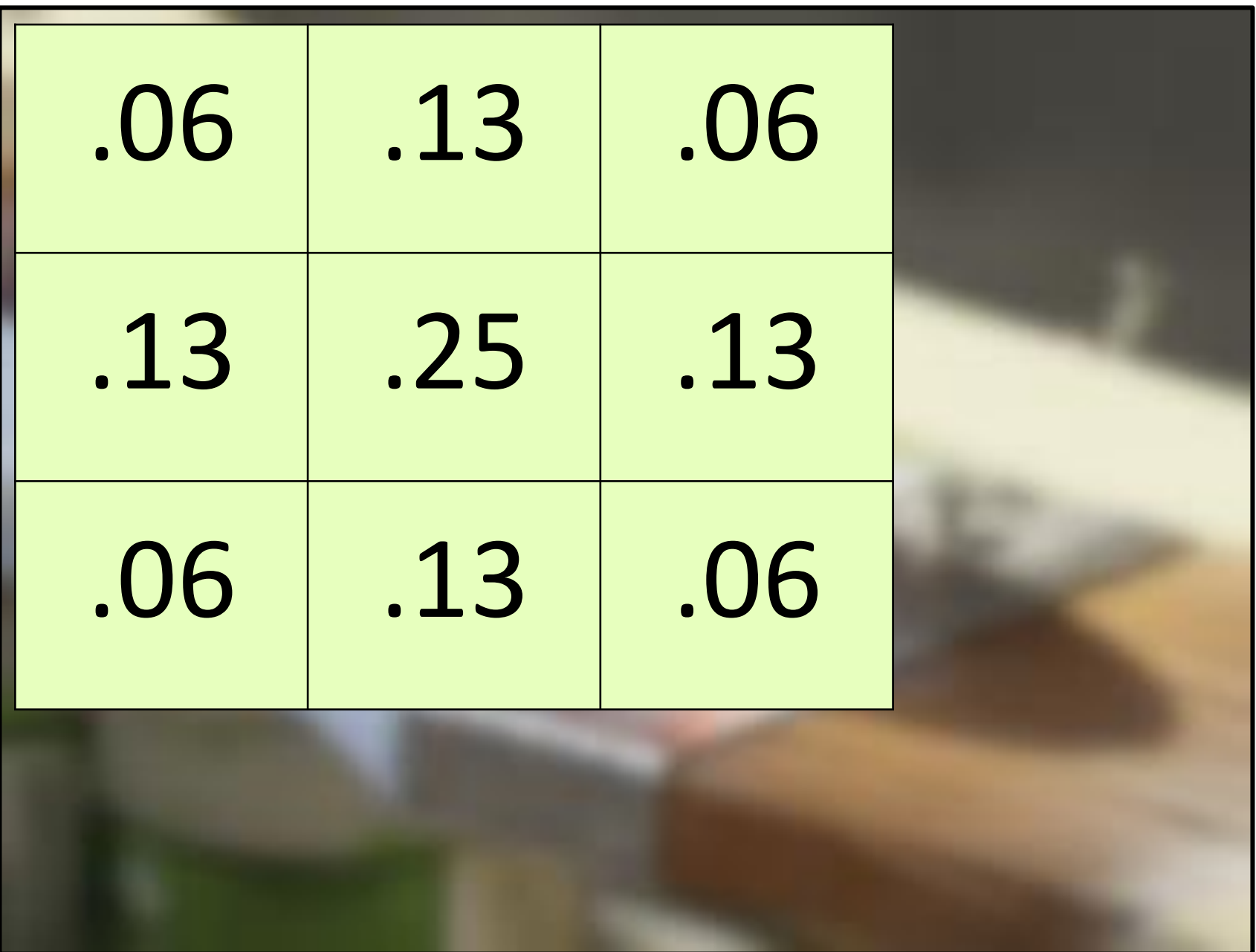


Original Image



Blur

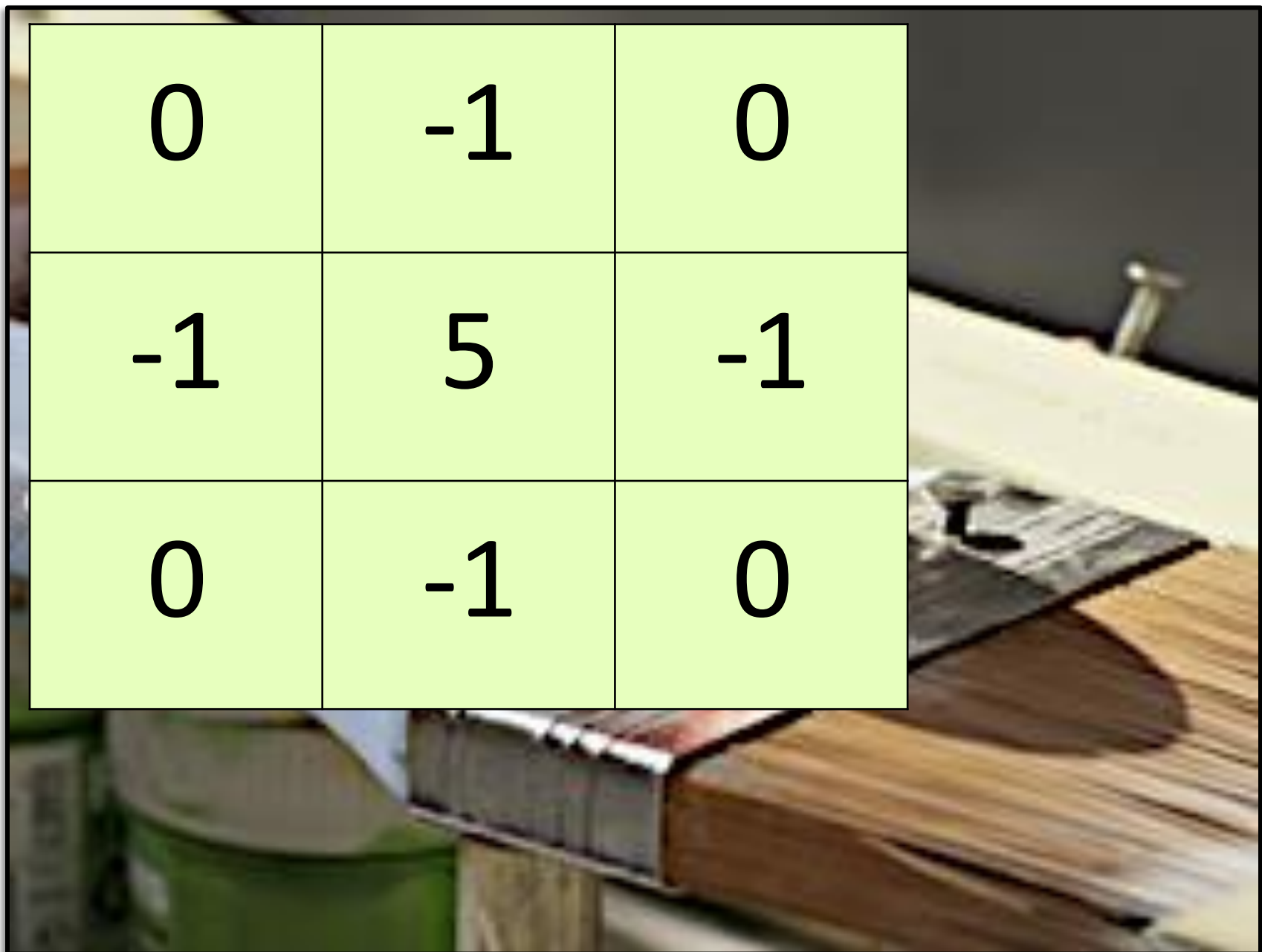
.06	.13	.06
.13	.25	.13
.06	.13	.06

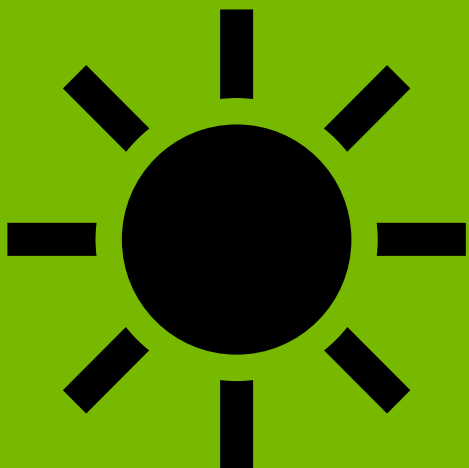




Sharpen

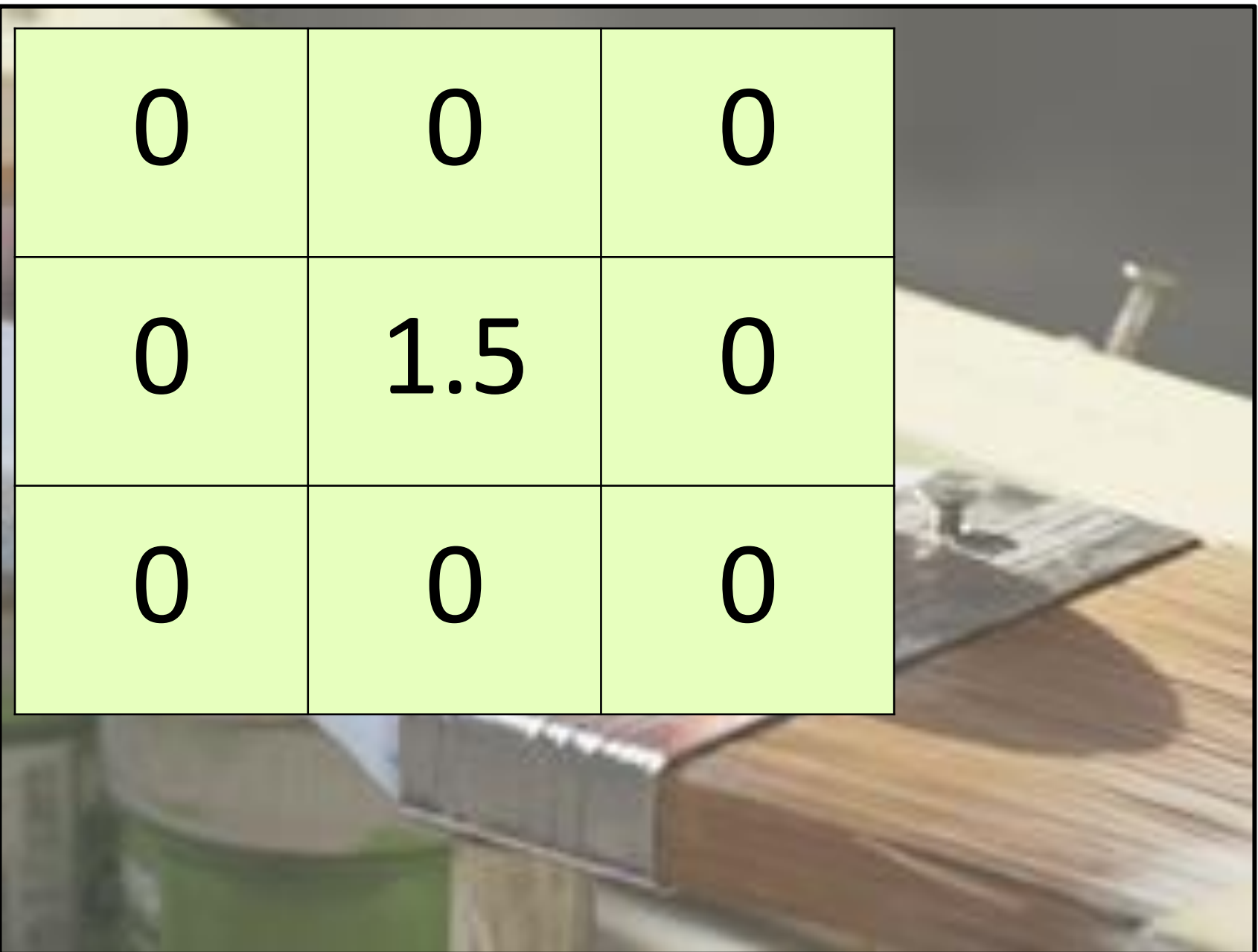
0	-1	0
-1	5	-1
0	-1	0





Brighten

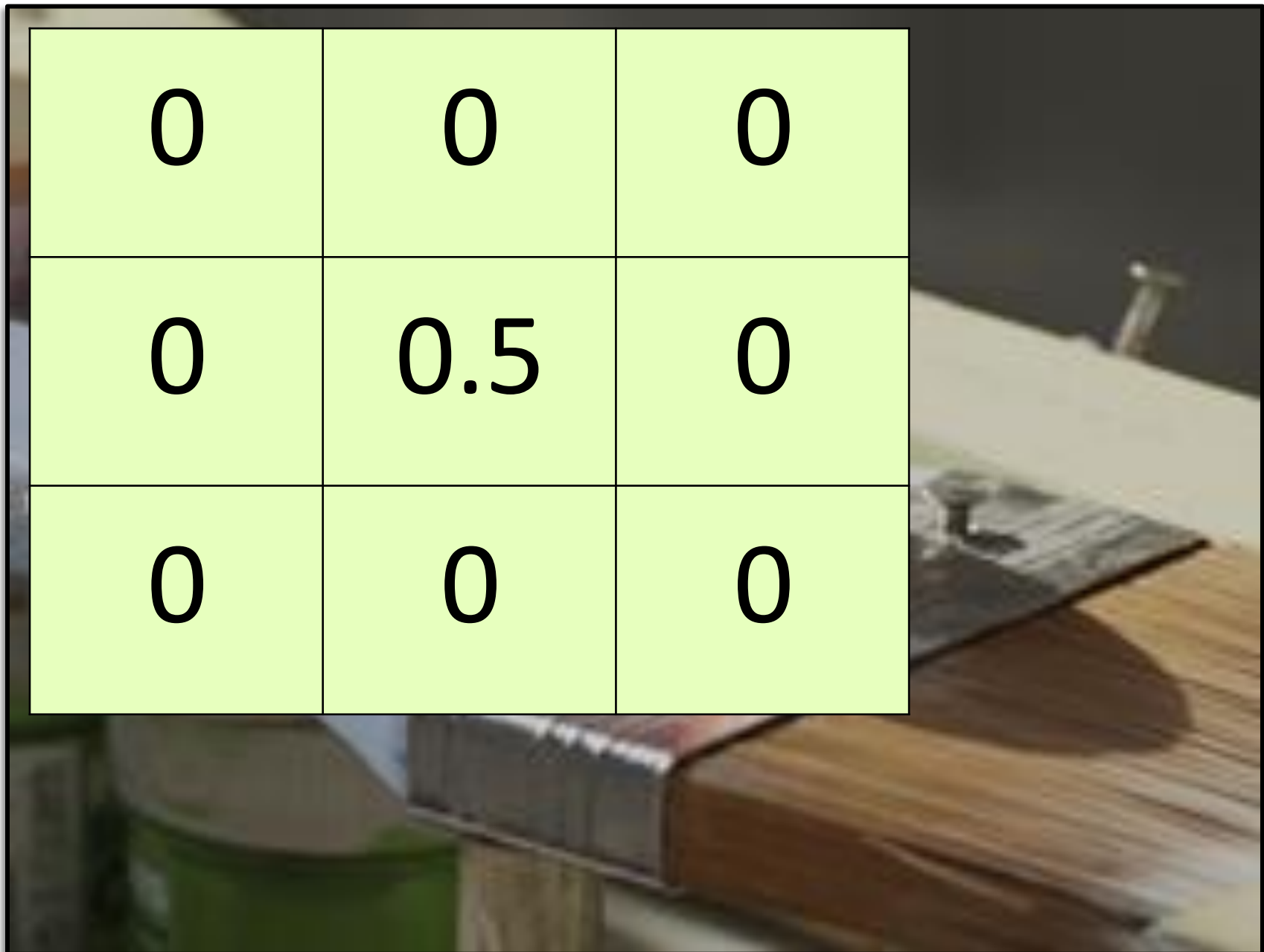
0	0	0
0	1.5	0
0	0	0





Darken

0	0	0
0	0.5	0
0	0	0



Kernels and Convolution

Blur Kernel

.06	.13	.06
.13	.25	.13
.06	.13	.06

*

Original Image

1	0	1	1	0	1
0	1	0	0	1	0
0	1	1	1	1	0
0	1	1	1	1	0
1	0	1	1	0	1
1	1	0	0	1	1

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Convolved Image

Kernels and Convolution

Blur Kernel

.06	.13	.06
.13	.25	.13
.06	.13	.06

*

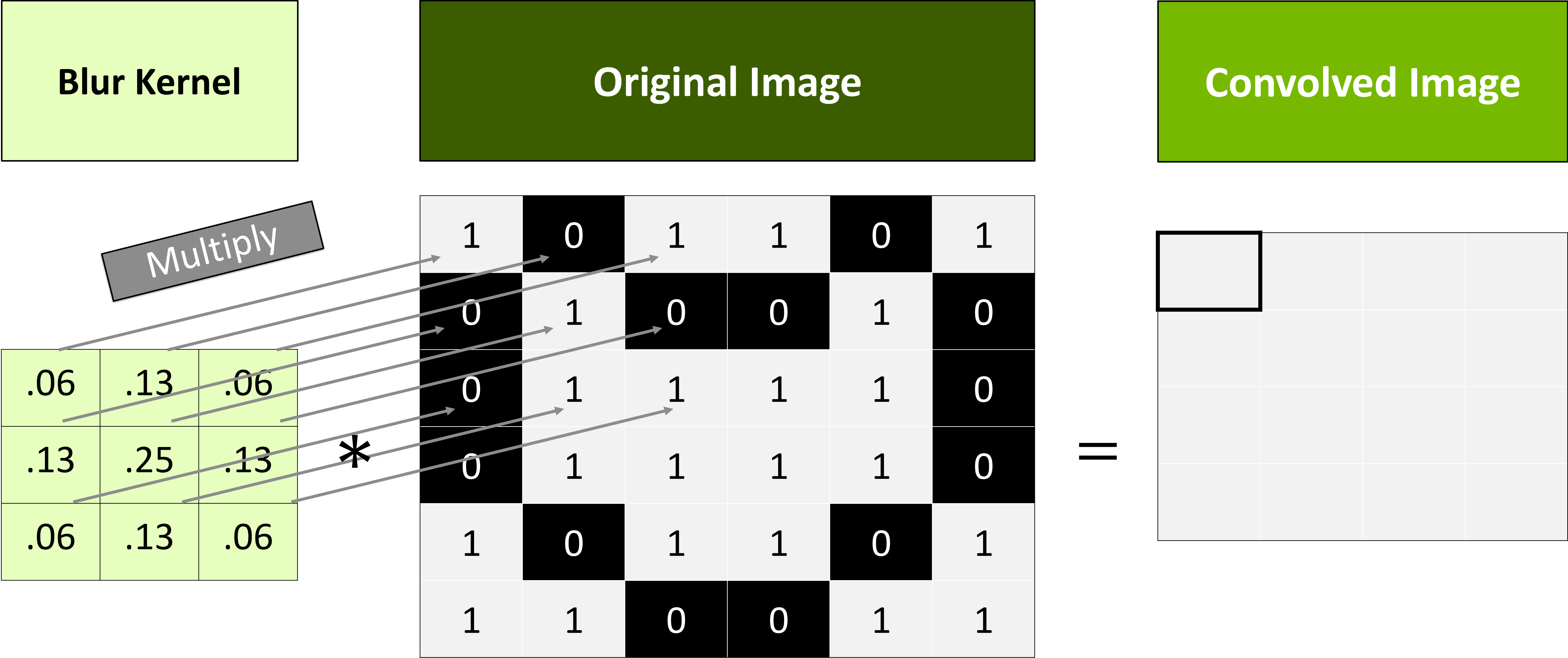
Original Image

1	0	1	1	0	1
0	1	0	0	1	0
0	1	1	1	1	0
0	1	1	1	1	0
1	0	1	1	0	1
1	1	0	0	1	1

=

Convolved Image

Kernels and Convolution



Kernels and Convolution

Blur Kernel

Original Image

Convolved Image

.06	.13	.06
.13	.25	.13
.06	.13	.06

*

.06	0	.06	1	0	1
0	.25	0	Total		0
0	.13	.06	1	1	0
0	1	1	1	1	0
1	0	1	1	0	1
1	1	0	0	1	1

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.56			

Kernels and Convolution

Blur Kernel

.06	.13	.06
.13	.25	.13
.06	.13	.06

*

Original Image

1	0	.13	.06	0	1
0	.13	0	0	1	0
0	.06	.13	.06	1	0
0	1	1	1	1	0
1	0	1	1	0	1
1	1	0	0	1	1

=

Convolved Image

.56	.57		

Kernels and Convolution

Blur Kernel

.06	.13	.06
.13	.25	.13
.06	.13	.06

*

Original Image

1	0	1	1	0	1
0	1	0	0	1	0
0	1	1	1	1	0
0	1	1	1	1	0
1	0	1	1	0	1
1	1	0	0	1	1

=

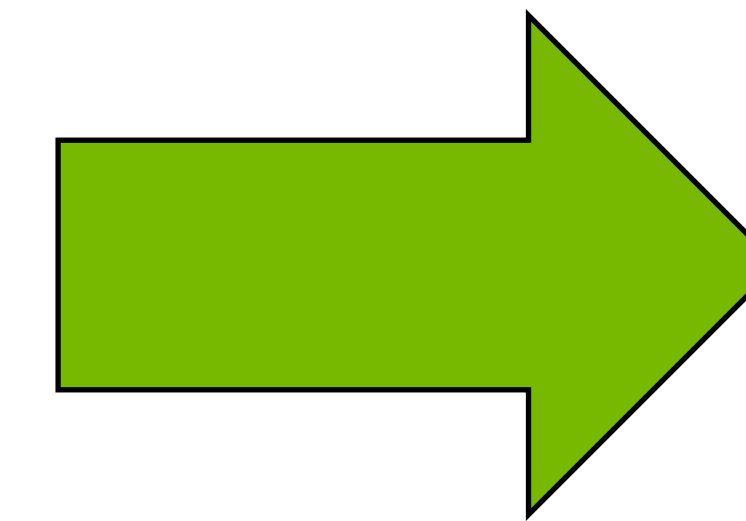
Convolved Image

.56	.57	.57	.56
.7	.82	.82	.7
.69	.95	.95	.69
.64	.69	.69	.64

Stride

Stride 1

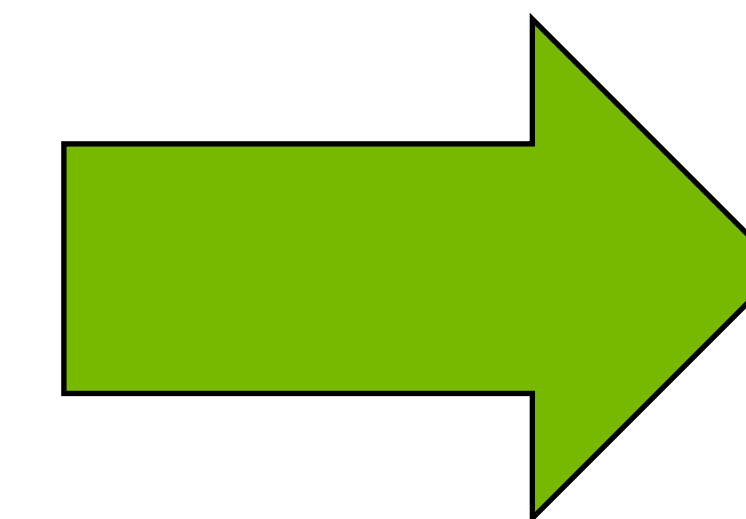
1	0	1	1	0	1
0	1	0	0	1	0
0	1	1	1	1	0



.56	.57	.57	.56
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Stride 2

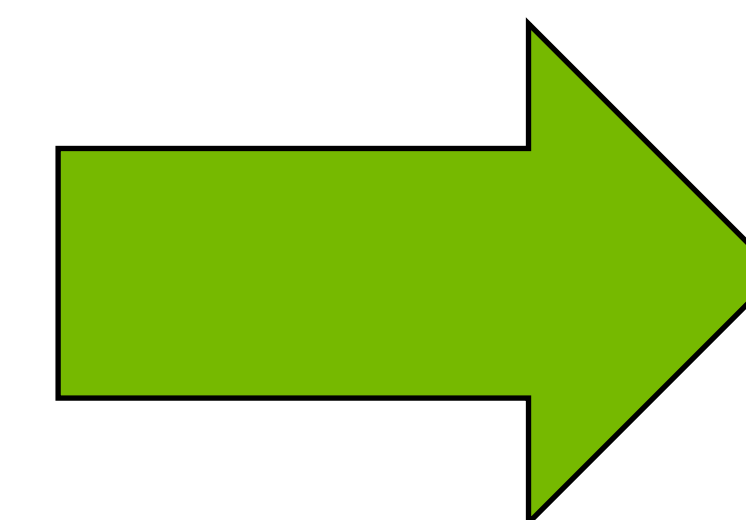
1	0	1	1	0	1
0	1	0	0	1	0
0	1	1	1	1	0



.56	.57
-----	-----

Stride 3

1	0	1	1	0	1
0	1	0	0	1	0
0	1	1	1	1	0



.56	.56
-----	-----

Padding

Original Image

1	0	1	1	0	1
0	1	0	0	1	0
0	1	1	1	1	0
0	1	1	1	1	0
1	0	1	1	0	1
1	1	0	0	1	1

Zero Padding

0	0	0	0	0	0	0	0
0	1	0	1	1	0	1	0
0	0	1	0	0	1	0	0
0	0	1	1	1	1	0	0
0	0	1	1	1	1	0	0
0	1	0	1	1	0	1	0
0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	0

Padding

Original Image

1	0	1	1	0	1
0	1	0	0	1	0
0	1	1	1	1	0
0	1	1	1	1	0
1	0	1	1	0	1
1	1	0	0	1	1

Mirror Padding

1	1	0	1	1	0	1	1
1	1	0	1	1	0	1	1
0	0	1	0	0	1	0	0
0	0	1	1	1	1	0	0
0	0	1	1	1	1	0	0
1	1	0	1	1	0	1	1
1	1	1	0	0	1	1	1
1	1	1	0	0	1	1	1



Kernels and Neural Networks

Kernels and Neural Networks

Kernel

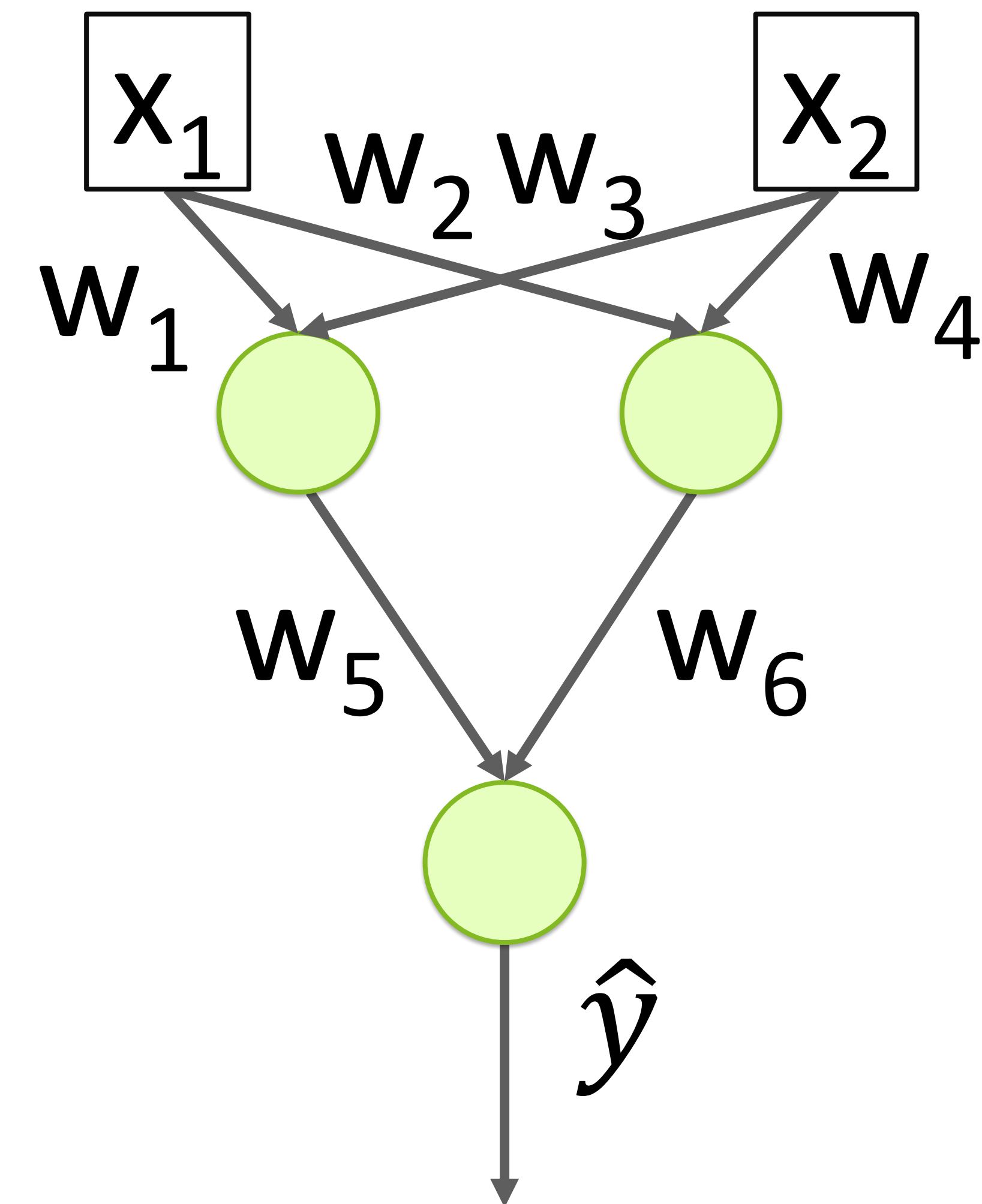
w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

Kernels and Neural Networks

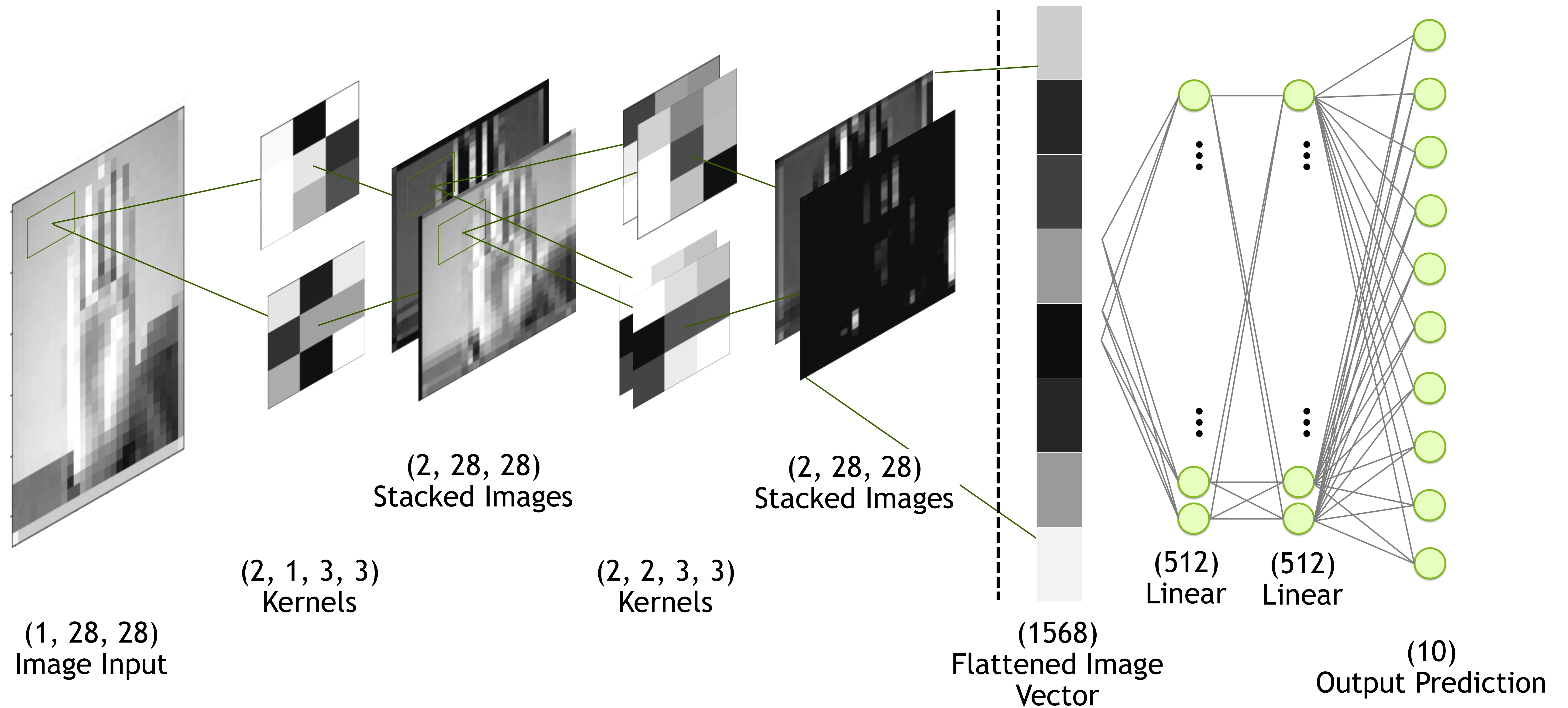
Kernel

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

Neuron



Kernels and Neural Networks



Finding Edges

Vertical Edges



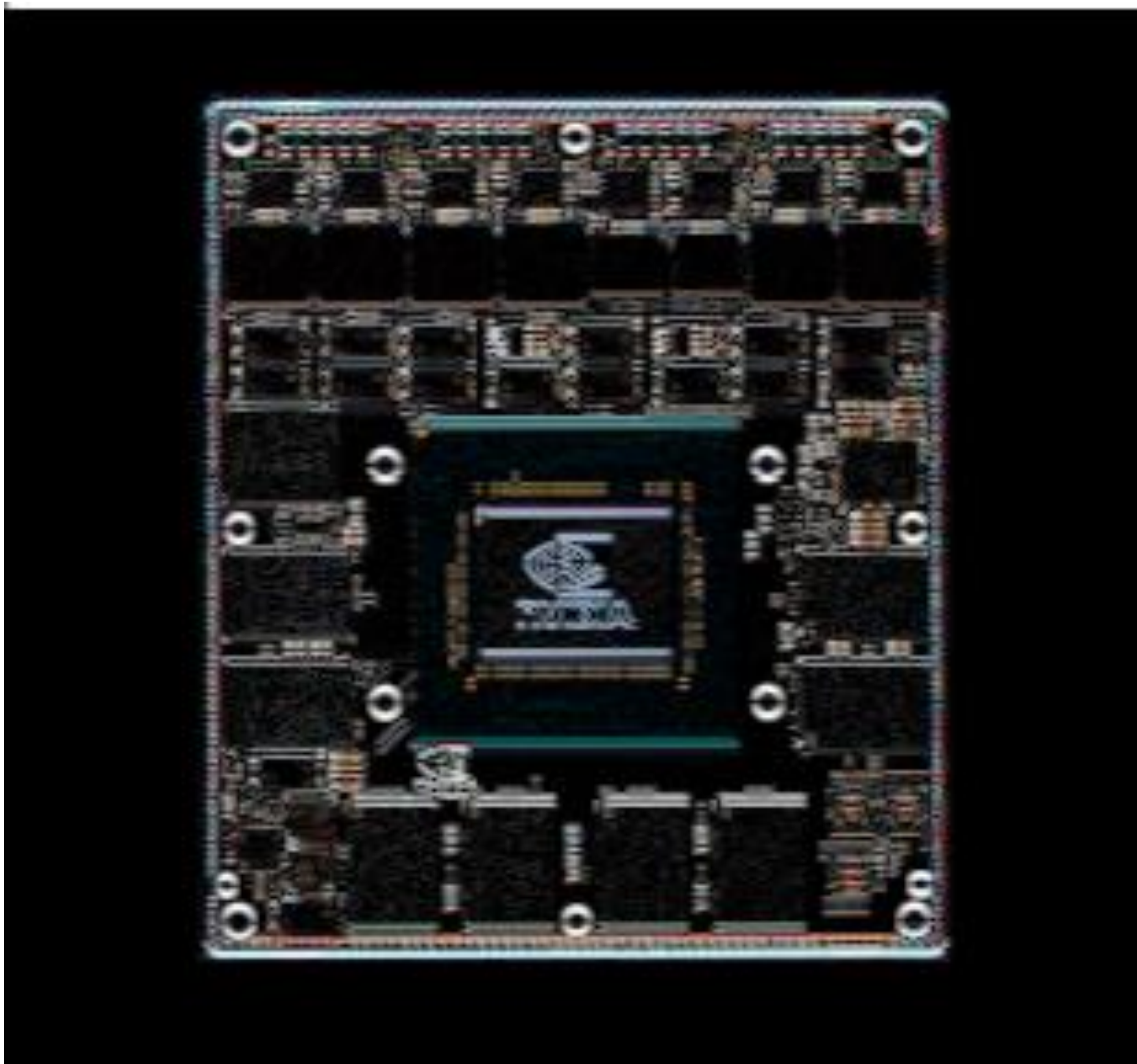
1	0	-1
2	0	-2
1	0	-1

Original Image



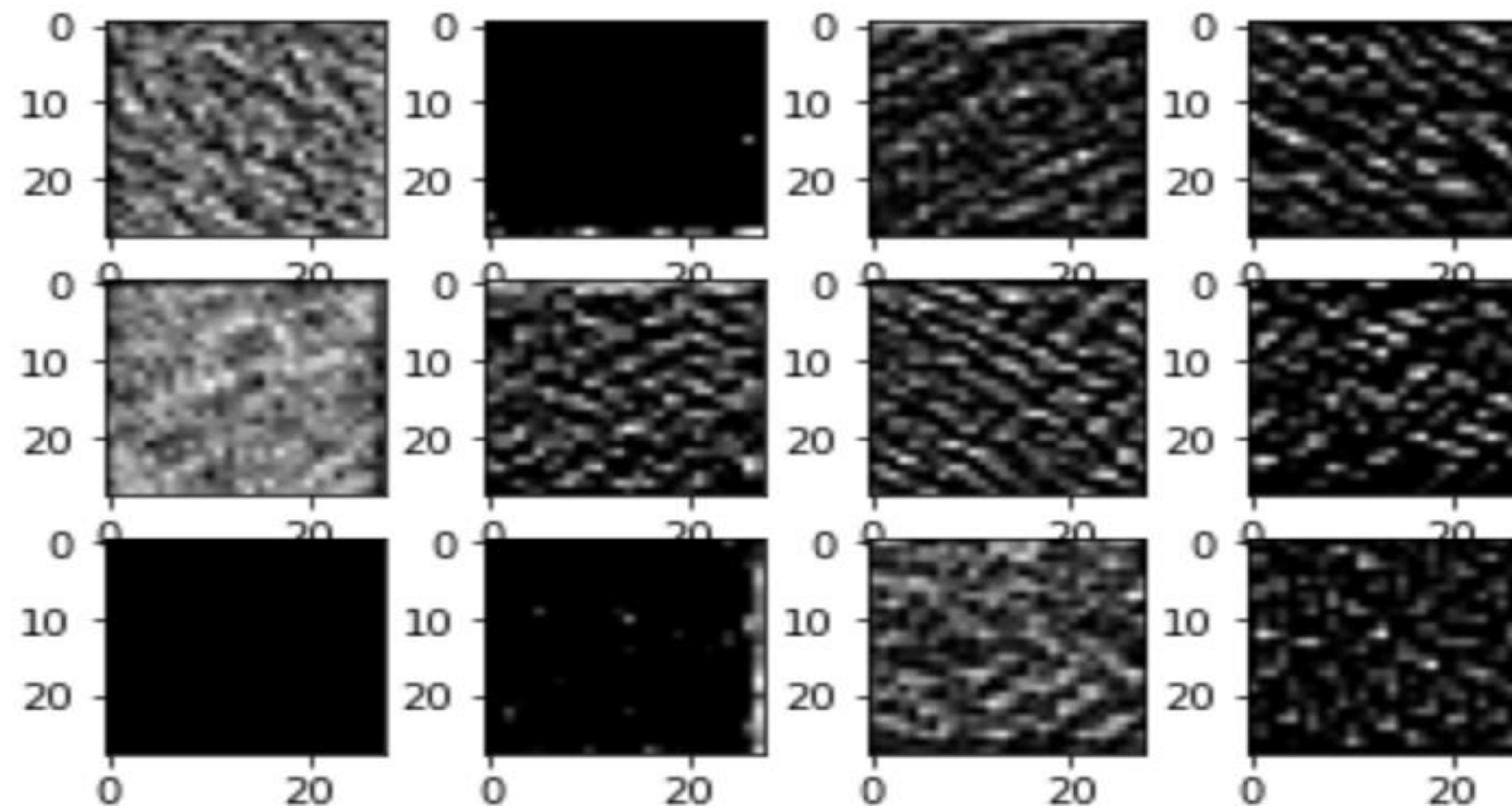
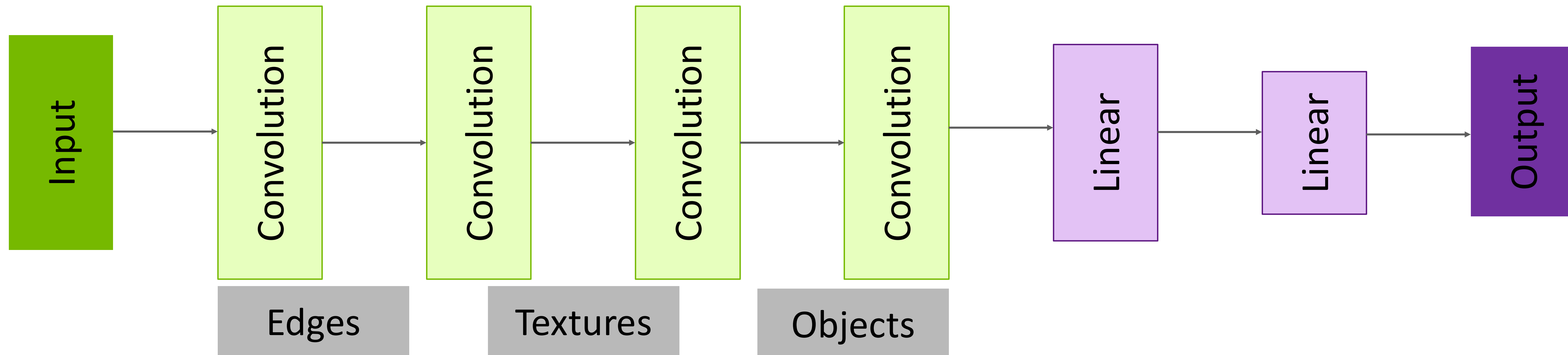
0	0	0
0	1	0
0	0	0

Horizontal Edges



1	2	1
0	0	0
-1	-2	-1

Neural Network Perception



Neural Network Perception

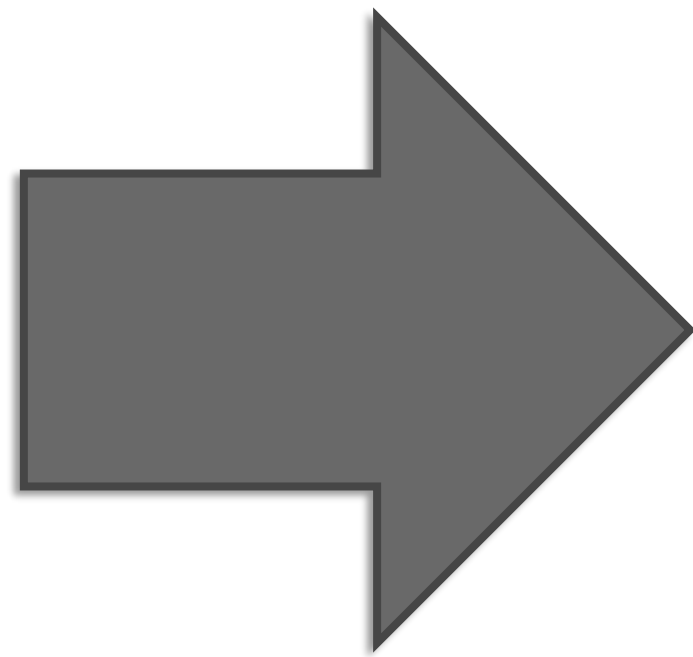




Other Layers in the Model

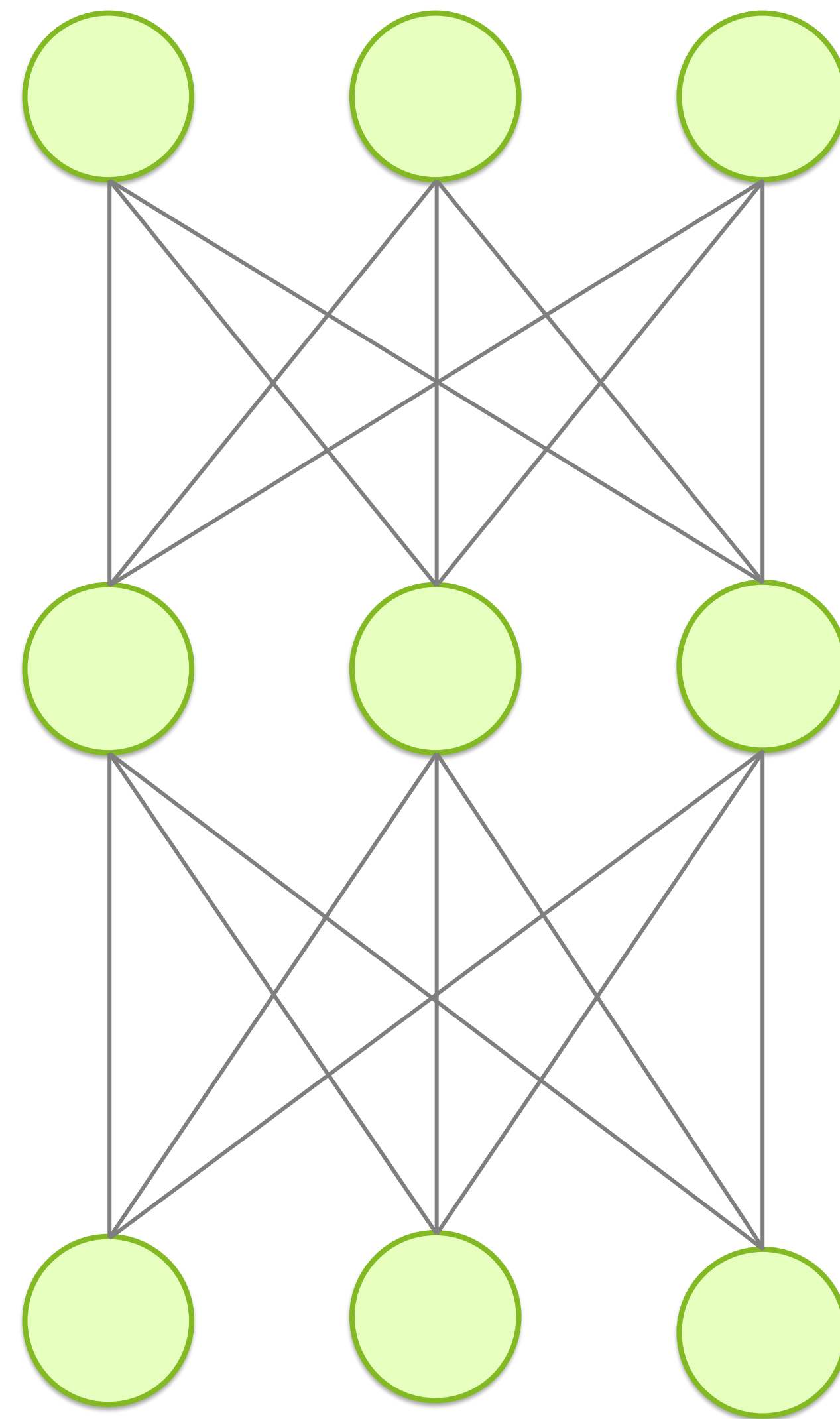
Max Pooling

110	256	153	67
12	89	88	43
10	15	50	55
23	9	49	23

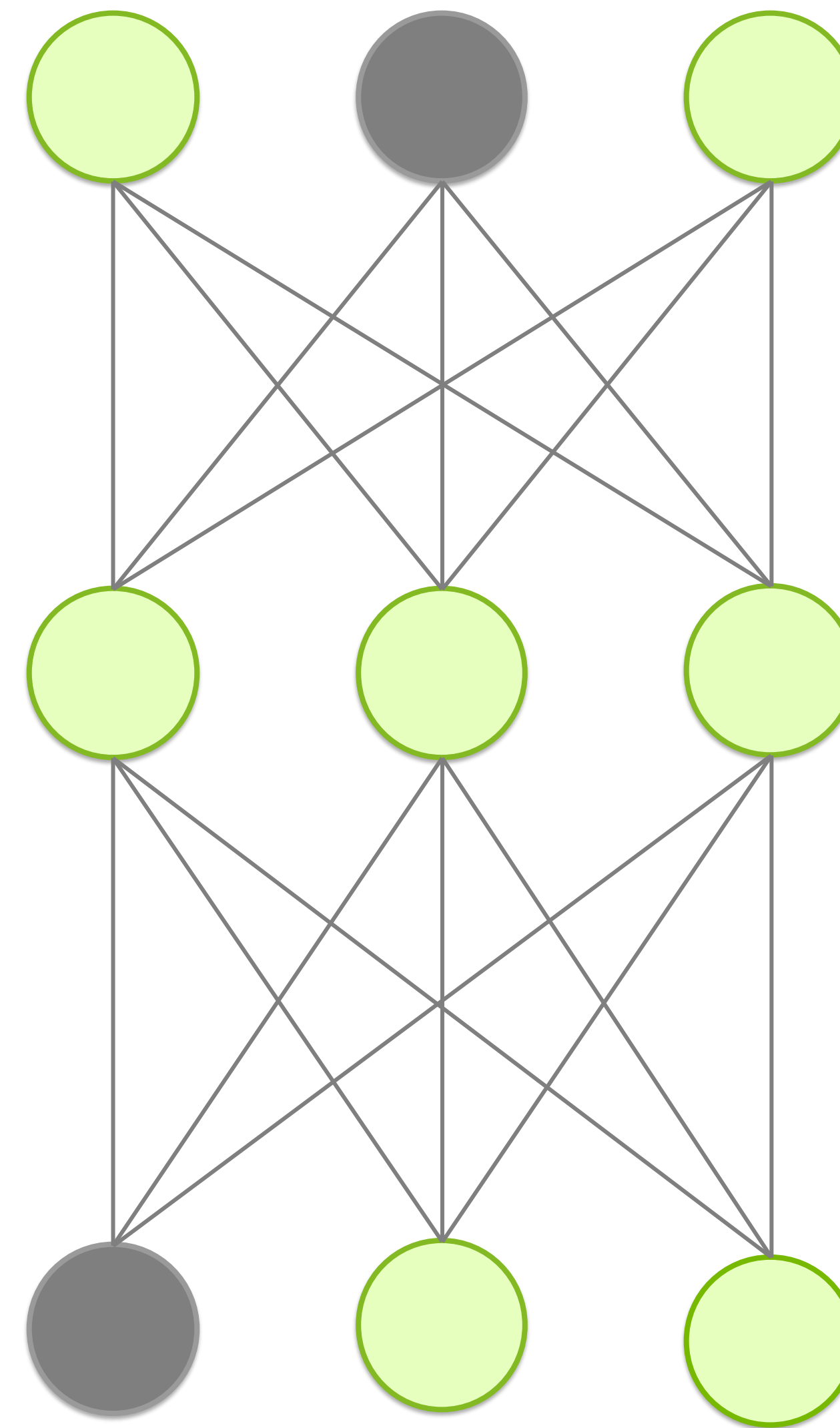


256	153
23	55

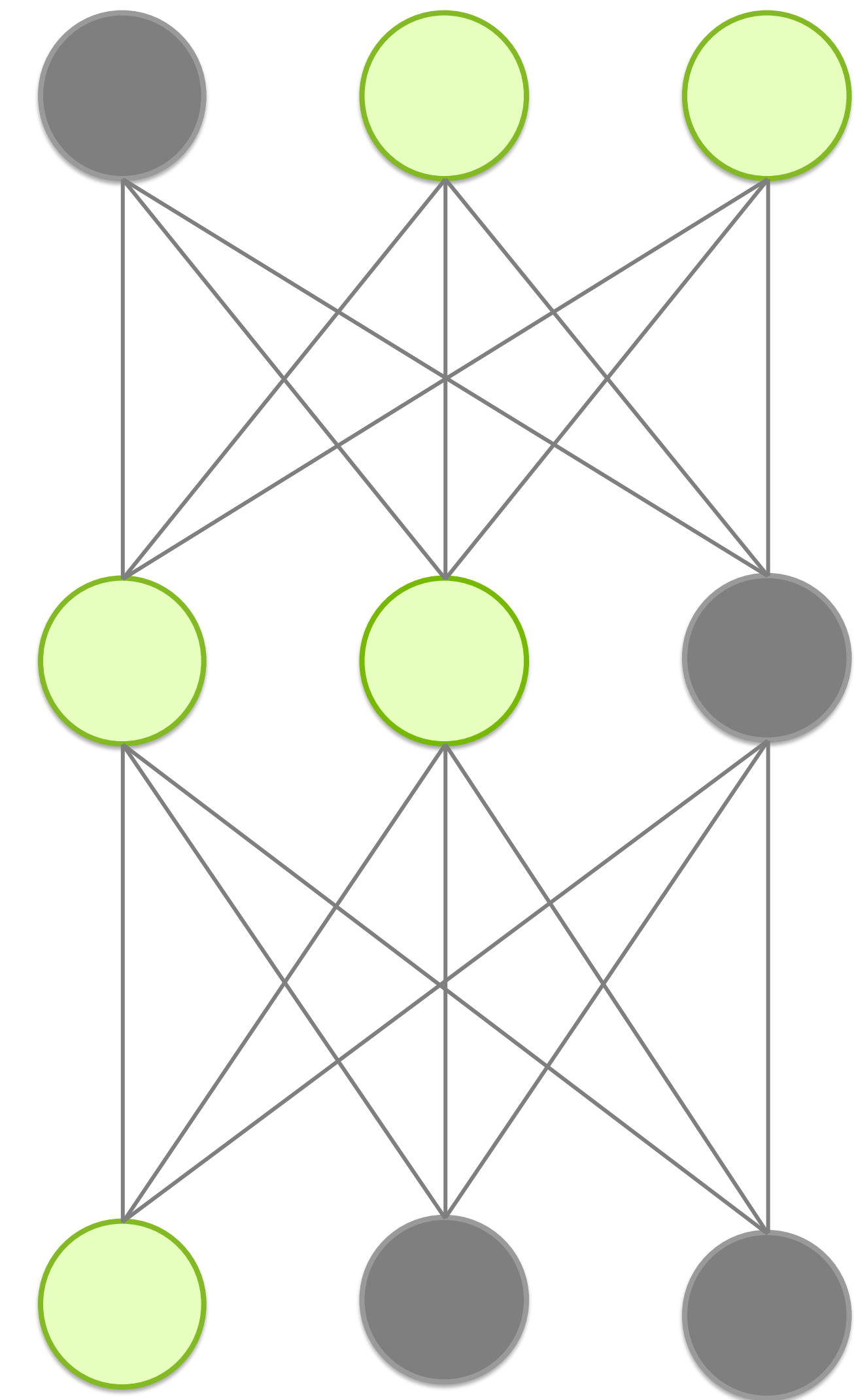
Dropout



rate = 0

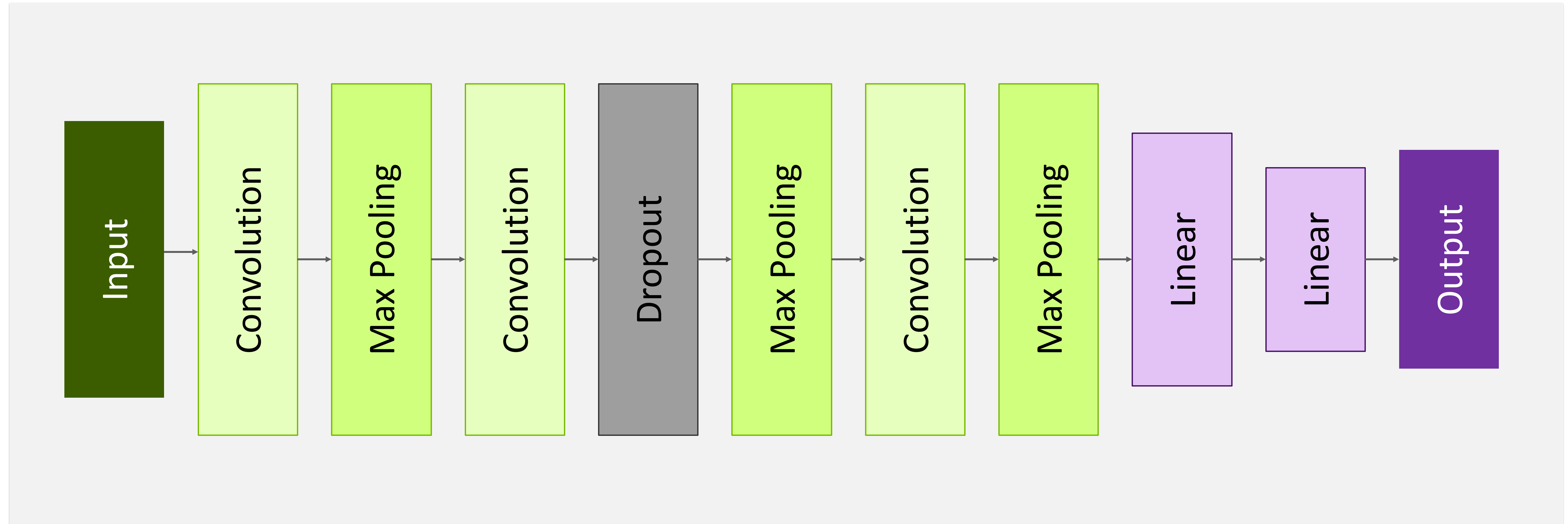


rate = .2



rate = .4

Whole Architecture





Let's go!

